

```
import mysql.connector

from mysql.connector import Error


# =====

# NUCLEUS OS

# Nuclear Waste & Radiation Monitoring System

# Python + MySQL Console Application

# =====


DB_CONFIG = {
    "host": "localhost",
    "user": "root",
    "password": "YOUR_MYSQL_PASSWORD",
    "database": "nucleus_os"
}


# =====

# DATABASE CONNECTION

# =====


def connect_database():

    try:
        conn = mysql.connector.connect(
            host=DB_CONFIG["host"],
            user=DB_CONFIG["user"],
            password=DB_CONFIG["password"],
```

```
        database=DB_CONFIG["database"]
    )
```

```
    return conn
```

```
except Error as e:
```

```
    print("\nDatabase connection failed!")
    print("Error:", e)
    return None
```

```
# =====
# DISPLAY TABLE
# =====
```

```
def display_table(conn, table, columns):
```

```
    try:
```

```
        cursor = conn.cursor()
```

```
        query = f"SELECT {' , '.join(columns)} FROM {table}"
```

```
        cursor.execute(query)
```

```
        rows = cursor.fetchall()
```

```
        print("\n" + "=" * 90)
```

```
print(table.upper())
```

```
print("=" * 90)
```

```
if not rows:
```

```
    print("No records found.")
```

```
    cursor.close()
```

```
    return
```

```
print(" | ".join(columns))
```

```
print("-" * 90)
```

```
for row in rows:
```

```
    print(" | ".join(str(value) for value in row))
```

```
cursor.close()
```

```
except Error as e:
```

```
    print("Database Error:", e)
```

```
# =====
```

```
# ADD RECORD
```

```
# =====
```

```
def add_record(conn, table, columns):
```

```
    values = []
```

```
print("\nEnter the following details:")
```

```
for column in columns:
```

```
    value = input(
        column.replace("_", " ").title() + ": "
    )
```

```
    if value == "":
        print("All fields are compulsory.")
        return
```

```
    values.append(value)
```

```
try:
```

```
    cursor = conn.cursor()
```

```
    placeholders = ",".join(["%s"] * len(values))
```

```
    query = (
        f"INSERT INTO {table} "
        f"({','.join(columns)}) "
        f"VALUES ({placeholders})"
    )
```

```
    cursor.execute(query, values)
```

```
    conn.commit()
```

```
cursor.close()
```

```
print("\nRecord added successfully!")
```

```
except Error as e:
```

```
conn.rollback()
```

```
print("\nDatabase Error:", e)
```

```
# =====
```

```
# DELETE RECORD
```

```
# =====
```

```
def delete_record(conn, table, primary_key):
```

```
    record_id = input(
        f"\nEnter {primary_key.replace('_', ' ')} to delete: "
    )
```

```
    try:
```

```
        cursor = conn.cursor()
```

```
        query = (
            f"DELETE FROM {table} "
            f"WHERE {primary_key}=%s"
```

)

cursor.execute(query, (record_id,))

conn.commit()

if cursor.rowcount > 0:

print("\nRecord deleted successfully!")

else:

print("\nRecord not found.")

cursor.close()

except Error as e:

conn.rollback()

print("\nDatabase Error:", e)

=====

GENERIC MODULE

=====

def database_module(

conn,

title,

table,

columns,

```
input_columns,  
primary_key  
):
```

```
while True:
```

```
    print("\n")  
    print("=" * 60)  
    print(title.upper())  
    print("=" * 60)
```

```
    print("1. View Records")  
    print("2. Add Record")  
    print("3. Delete Record")  
    print("4. Back")
```

```
    choice = input("\nEnter your choice: ")
```

```
    if choice == "1":
```

```
        display_table(  
            conn,  
            table,  
            columns  
        )
```

```
    elif choice == "2":
```

```
        add_record(  
            conn,  
            table,  
            columns,  
            primary_key,  
            input_columns,  
            input_values  
        )
```

```
        conn,
        table,
        input_columns
    )
```

```
elif choice == "3":
```

```
    delete_record(
        conn,
        table,
        primary_key
    )
```

```
elif choice == "4":
```

```
    break
```

```
else:
```

```
    print("\nInvalid choice!")
```

```
# =====
# DASHBOARD
# =====
```

```
def dashboard(conn, username):
```

```
    while True:
```

```
print("\n")
print("=" * 70)
print("          NUCLEUS OS")
print("    Nuclear Waste & Radiation Monitoring System")
print("=" * 70)
```

```
print(f"\nWelcome, {username}")
```

```
print("\n----- DASHBOARD -----")
```

```
print(
    "Nuclear Plants    :",
    count_records(conn, "nuclear_power_plants")
)
```

```
print(
    "Reactors          :",
    count_records(conn, "reactors")
)
```

```
print(
    "Employees         :",
    count_records(conn, "employees")
)
```

```
print(
    "Radiation Sensors  :",
    count_records(conn, "radiation_monitoring")
)
```

)

```
print(
    "Storage Facilities  :",
    count_records(conn, "waste_storage")
)
```

```
print(
    "Safety Records    :",
    count_records(conn, "safety_records")
)
```

```
print("\n----- MAIN MENU -----")
```

```
print("1. Users / Login")
print("2. Nuclear Power Plant")
print("3. Reactors")
print("4. Power Generation")
print("5. Waste Storage")
print("6. Radiation Monitoring")
print("7. HR & Employees")
print("8. Radiation Exposure")
print("9. Safety Management")
print("10. Career Protocol")
print("11. Payroll")
print("12. Logout")
```

```
choice = input("\nEnter your choice: ")
```

```
# -----
```

```
# USERS
```

```
# -----
```

```
if choice == "1":
```

```
    database_module(
```

```
        conn,
```

```
        "Users",
```

```
        "users",
```

```
        [
```

```
            "user_id",
```

```
            "username",
```

```
            "password",
```

```
            "email"
```

```
        ],
```

```
        [
```

```
            "username",
```

```
            "password",
```

```
            "email"
```

```
        ],
```

```
        "user_id"
```

```
    )
```

```
# -----
```

```
# NUCLEAR POWER PLANT
```

```
# -----
```

```
elif choice == "2":
```

```
database_module(  
    conn,  
    "Nuclear Power Plant",  
    "nuclear_power_plants",  
    [  
        "plant_id",  
        "plant_name",  
        "location"  
    ],  
    [  
        "plant_name",  
        "location"  
    ],  
    "plant_id"  
)
```

```
# -----
```

```
# REACTORS
```

```
# -----
```

```
elif choice == "3":
```

```
database_module(  
    conn,  
    "Reactors",  
    "reactors",  
    [  
        "reactor_id",
```

```
        "plant_id",
        "reactor_type",
        "capacity",
        "operating_status"
    ],
    [
        "plant_id",
        "reactor_type",
        "capacity",
        "operating_status"
    ],
    "reactor_id"
)
```

```
# -----
```

```
# POWER GENERATION
```

```
# -----
```

```
elif choice == "4":
```

```
database_module(
    conn,
    "Power Generation",
    "power_generation",
    [
        "generation_id",
        "reactor_id",
        "generation_date",
        "electricity_generated"
```

```
],  
[  
    "reactor_id",  
    "generation_date",  
    "electricity_generated"  
],  
"generation_id"  
)
```

```
# -----
```

```
# WASTE STORAGE
```

```
# -----
```

```
elif choice == "5":
```

```
database_module(  
    conn,  
    "Waste Storage",  
    "waste_storage",  
    [  
        "storage_id",  
        "storage_building",  
        "capacity",  
        "available_space",  
        "inspection_records"  
    ],  
    [  
        "storage_building",  
        "capacity",
```

```
        "available_space",
        "inspection_records"
    ],
    "storage_id"
)
```

```
# -----
```

```
# RADIATION MONITORING
```

```
# -----
```

```
elif choice == "6":
```

```
    database_module(
        conn,
        "Radiation Monitoring",
        "radiation_monitoring",
        [
            "sensor_record_id",
            "sensor_name",
            "radiation_reading",
            "alert_generation",
            "area_monitoring",
            "recorded_at"
        ],
        [
            "sensor_name",
            "radiation_reading",
            "alert_generation",
            "area_monitoring"
        ]
    )
```

```
],  
    "sensor_record_id"  
)
```

```
# -----  
# EMPLOYEES  
# -----
```

```
elif choice == "7":
```

```
    database_module(  
        conn,  
        "HR & Employees",  
        "employees",  
        [  
            "employee_id",  
            "employee_name",  
            "department",  
            "salary",  
            "promotion_status",  
            "performance_record"  
        ],  
        [  
            "employee_name",  
            "department",  
            "salary",  
            "promotion_status",  
            "performance_record"  
        ],  
    )
```

```
        "employee_id"
    )

# -----
# EXPOSURE
# -----

elif choice == "8":

    database_module(
        conn,
        "Employee Radiation Exposure",
        "employee_exposure",
        [
            "exposure_id",
            "employee_id",
            "daily_exposure",
            "monthly_exposure",
            "safe_limit",
            "limit_status",
            "recorded_date"
        ],
        [
            "employee_id",
            "daily_exposure",
            "monthly_exposure",
            "safe_limit",
            "limit_status",
            "recorded_date"
```

```
],  
    "exposure_id"  
)
```

```
# -----  
# SAFETY  
# -----
```

```
elif choice == "9":
```

```
    database_module(  
        conn,  
        "Safety Management",  
        "safety_records",  
        [  
            "safety_id",  
            "safety_title",  
            "safety_description",  
            "risk_level",  
            "action_required",  
            "created_at"  
        ],  
        [  
            "safety_title",  
            "safety_description",  
            "risk_level",  
            "action_required"  
        ],  
        "safety_id"
```

)

CAREER

elif choice == "10":

database_module(

conn,

"Career Protocol",

"career_records",

[

"career_id",

"employee_id",

"position_name",

"candidate_name",

"career_status",

"notes"

],

[

"employee_id",

"position_name",

"candidate_name",

"career_status",

"notes"

],

"career_id"

)

```
# -----
```

```
# PAYROLL
```

```
# -----
```

```
elif choice == "11":
```

```
    database_module(  
        conn,  
        "Payroll Management",  
        "payroll",  
        [  
            "payroll_id",  
            "employee_id",  
            "salary",  
            "payment_date",  
            "payment_status"  
        ],  
        [  
            "employee_id",  
            "salary",  
            "payment_date",  
            "payment_status"  
        ],  
        "payroll_id"  
    )
```

```
# -----
```

```
# LOGOUT
```

```
# -----
```

```
elif choice == "12":
```

```
    print("\nLogged out successfully.")
```

```
    break
```

```
else:
```

```
    print("\nInvalid choice!")
```

```
# =====
```

```
# COUNT RECORDS
```

```
# =====
```

```
def count_records(conn, table):
```

```
    try:
```

```
        cursor = conn.cursor()
```

```
        cursor.execute(
```

```
            f"SELECT COUNT(*) FROM {table}"
```

```
        )
```

```
        result = cursor.fetchone()[0]
```

```
        cursor.close()
```

```
return result
```

```
except Error:
```

```
return 0
```

```
# =====
```

```
# LOGIN
```

```
# =====
```

```
def login(conn):
```

```
    print("\n")
```

```
    print("=" * 60)
```

```
    print("          NUCLEUS OS")
```

```
    print("=" * 60)
```

```
    print("\nNuclear Waste & Radiation Monitoring System")
```

```
    username = input("\nUsername : ")
```

```
    password = input("Password : ")
```

```
    if username == "" or password == "":
```

```
        print("\nUsername and password are compulsory.")
```

```
        return
```

try:

```
cursor = conn.cursor()
```

```
query = """
```

```
SELECT user_id, username, email
```

```
FROM users
```

```
WHERE username=%s AND password=%s
```

```
"""
```

```
cursor.execute(
```

```
    query,
```

```
    (username, password)
```

```
)
```

```
user = cursor.fetchone()
```

```
cursor.close()
```

```
if user:
```

```
    print("\nLogin successful!")
```

```
    dashboard(
```

```
        conn,
```

```
        user[1]
```

```
    )
```

```
else:
```

```
print("\nInvalid username or password.")
```

```
except Error as e:
```

```
print("\nDatabase Error:", e)
```

```
# =====
```

```
# MAIN PROGRAM
```

```
# =====
```

```
def main():
```

```
    print("=" * 60)
```

```
    print("          NUCLEUS OS STARTING...")
```

```
    print("=" * 60)
```

```
    conn = connect_database()
```

```
    if conn is None:
```

```
        return
```

```
    login(conn)
```

```
    conn.close()
```

```
    print("\nNucleus OS closed.")
```

```
    print("Thank you.")
```

```
# =====
```

```
# RUN PROGRAM
```

```
# =====
```

```
if __name__ == "__main__":
```

```
    main()
```